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Article

Seven Decades of Project Portfolio Management Research (1950–2019) and Perspectives for the Future

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Abstract

Although project portfolio management (PPM) has been on the research agenda for at least seven decades (1950–2019), the number of contributions on the topic continues to grow. Literature reviews help us to integrate and build on what we already know and to understand where we came from. However, reviews proving a shared historical narrative of the discipline are scarce but key. Thus, we trace the literature back to its origin, building and synthesizing our historical narrative into a model. We evaluate what has already happened and what will most likely shape the future.

Keywords

project portfolio management, historical analysis, narrative literature review, temporarity, governance, strategy-oriented portfolios, project as a practice, readiness to change

Introduction

As most countries and firms become increasingly *projectified* (Midler, 1995, 2019), organizations are experiencing ongoing challenges in managing their project portfolios, since projects constitute a major part of organizational budgets and strategic development (Schoper et al., 2018). Thus, organizations apply project portfolio management (PPM) to strategically and holistically manage project portfolios (Clegg et al., 2018). As a discipline, PPM owes its legacy to finance theory developed in the 1950s (Martin, 1955; W. T. Morris, 1958)—often associated with the Nobel Prize winning work by Harry Markowitz (Markowitz, 1952, 1991). Overall, portfolio theories can be characterized by their emphasis on the coordination of different entities, so their joint investment risks are reduced. The portfolio benefits and usage and development of scarce resources are optimized, and the portfolio stakeholders are considered appropriately (Srivastava & Gnyawali, 2011). These characteristics of theory on portfolios are not specific to temporary engagements like investments but can also be applied to enduring systems. Moreover, portfolio theory can be used to manage all kinds of investments and activities, for example, strategic alliance portfolios (Wassmer, 2008) and relationship portfolios (Tikkanen et al., 2007).

One ongoing challenge for the theory is to demonstrate its usefulness in practice (Gear, 1974). This is due to its complexity and its abstract principles. In that regard, since the 1960s, research and development (R&D) literature and new product development (NPD) literature have made an admirable effort to connect portfolio theory to the practical applications of PPM and accumulate this knowledge in literature reviews (Baker & Pound, 1964; Killen et al., 2007; Nobelius, 2004; Schmidt & Freeland, 1992). In particular, the work on portfolio management in NPD by Robert Cooper, Elko Kleinschmidt, and Scott Edgett—in the 1990s and 2000s—has pioneered and shaped the PPM discipline (Cooper et al., 2001).

During the last 15 to 20 years, the literature in the project management journals has gradually caught up and outpaced NPD literature regarding the project portfolio aspect (Hobbs, 2012). PPM has progressively grown as a research discipline with independent terms and theory (Enoch & Labuschagne, 2014; Martinsuo, 2013). Fruitful suggestions for the discipline's future development have been made (Clegg et al., 2018; Martinsuo & Geraldi, 2020); however, the discipline is in a nascent state when it comes to using the full potential of literature reviews as a research methodology. Meifort (2016) was indeed one of the few doing a structured review, but the contributions by Rothwell (1994) and Nobelius (2004) were more visionary in trying to understand and predict future development paths. Nobelius (2004) also stressed the importance of radical innovation projects that disrupt existing business ecologies and create new ones. Our article will provide some

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suggestions on what this means for future project portfolio management—as we have now been living for more than 15 years in an environment of open innovation, disruption, and business ecologies.

Furthermore, despite the effort by Rothwell (1994) and Nobelius (2004) to outline different generations of portfolio management, we found no contributions using the potential of developing a historical narrative based on the literature. This leaves us with the unaddressed question regarding the origin of the discipline's central concepts and what we can learn from the past to lay out a future research agenda. Current scholars emphasize the value of such historical analyses in organizational research (Gill et al., 2018); it sharpens one's vision of the present, not of the past, and stimulates thinking about alternative explanations (Van Fleet & Wren, 2005); it creates a baseline to evaluate how supposedly "new" practices are shaped (Thomson, 2001); and it enables us to learn the lessons of past mistakes to prepare for the future (Cummings & Bridgman, 2011). Finally, we believe that providing a shared historical narrative will strengthen the identity of the PPM discipline and motivate scholars and practitioners to continue its development.

Against this backdrop, our research aims to address the following research questions: "*What is the historical narrative of the PPM discipline, and what can we learn from the past to lay out a future research agenda*?"

We organize our article as follows: The next section describes the methodology, the steps taken in our structured literature review, and how we find and analyze our body of knowledge, consisting of 669 articles spread across seven decades and various research fields. The analysis section consists of two parts: a deconstruction section where we take the literature apart and a reconstruction section where we put the pieces together and form a historical narrative. Next, the discussion section synthesizes our narrative into a model and suggests a future agenda for the discipline by evaluating what has already happened and what will most likely shape the future. Finally, the last section presents our conclusions and the limitations of our research.

Methodology

Our historical analysis is organized as a narrative literature review (Baumeister & Leary, 1997), which requires extensive searching; reading; and tracing streams of literature back to the original sources by probing databases with keywords (Klein & Müller, 2020). Inspired by principles from structured literature review (SLR) methods (Denyer & Tranfield, 2009), we structured our review in three stages: (1) identifying keywords and developing a search string, (2) filtering articles, and (3) analyzing articles. To secure a high level of quality, we delimited our selection of articles to peer-reviewed academic journals. As an exception, we included articles from *Harvard Business Review* as this journal is a widely quoted scientific source with a major impact on the scientific discourse in management research (Schulz & Nicolai, 2015).

Identifying Keywords and Developing a Search String

To capitalize on what is already known, we started with an exploratory search for literature reviews and conceptual contributions on PPM. We found 24 relevant articles (see Appendix A), and we scanned these articles and used backward and forward tracking to identify the main keywords of the discipline (Webster & Watson, 2002). This process was insightful, but challenging, as the PPM articles' use of keywords and wording differ significantly across time, journals, and fields. In an iterative process, we utilized the identified keywords to develop the first version of our search string, and we evaluated its quality using two criteria. Criterion 1 was that the search string should have the ability to provide references from the entire period from 1950 to 2019 (i.e., the time span of interest for our historical analysis). As illustrated in Table 1, this required a range of subsearch strings in different databases and backward tracking to get a satisfactory result. Criterion 2 was that we wanted a benchmark to help us test whether our search results were likely to include the most relevant literature. For this purpose, we developed a benchmark consisting of 10 articles that we found *should* be included in the final search result due to their pronounced influence on the discipline. One proxy for the influence of a contribution is the number of citations (Webster & Watson, 2002). Thus, we did a second explorative search, this time in the Google Scholar (GS) database for PPM articles with more than 200 citations, which we determined to be a high number of citations in the PPM literature. The exploratory GS search resulted in 51 PPM articles. We randomly selected 10 papers from the 51 items and used these as our benchmark to evaluate the hit rate. The first test results using the search string only included seven of the 10 benchmark papers. Consequently, in various iterations, we refined our search strings and included more databases so that the final search result had all 10 benchmark papers. Finally, the search string conformed to the two defined quality criteria and resulted in 1,970 articles, and the next step was to filter these articles.

Filtering Articles

Filtering was performed by reviewing the titles and abstracts of 1,970 articles, which resulted in a final stock of 669 articles. In this process, we developed and utilized a set of inclusion criteria (see Appendix B). To increase the quality of the criteria for inclusion, we tracked intercoder agreement; this is defined as a measure of the extent to which independent judges make the same coding decisions (Lombard et al., 2006). Two researchers conducted the filtering process, until the criteria for inclusion were robust, by comparing the coding results and discussing discrepancies until an agreement was reached. The criteria for inclusion settled after an approximate filtering of 1,000 articles, and one researcher could do the filtering from this point.

Analyzing Data

Inspired by Torraco (2005), the analysis was conducted by *deconstructing* and *reconstructing* the literature. *Deconstruction* was performed by systematically (re)reading

Table 1. Identifying Keywords and Developing a Search String

Databases	Search Strings	Criterion 1: Time Span Covers the Period: 1950–2019	Criterion 2: Includes 10 Benchmark Papers	Search Results
Jstor	([Title: "Portfolio" OR Abstract: "Portfolio"] AND "project") OR ("Portfolio selection" <1960)	1950–1960	(1) Markowitz (1952)	89
Scopus (Search A)	Title: ("Portfolio") OR ("Multiproject") OR ("multi-project") AND (Abstract: "Portfolio" and "Project") OR (Abstract "Project studies")	–2019	(2) Cooper et al. (1997) (3) Archer and Ghasemzadeh (1999) (4) Martinsuo and Lehtonen (2007) (5) Blichfeldt and Eskerod (2008)	988
Scopus (Search B)	(Title: "Portfolio approach to Information systems") OR (Title: "IT portfolio management")	–2019	(6) McFarlan (1981) (7) Jeffery and Leliveld (2004)	32
Science Direct	Title + Abstract + Keywords: ("Project portfolio management")	–2019	(3) Archer and Ghasemzadeh (1999) (4) Martinsuo and Lehtonen (2007) (5) Blichfeldt and Eskerod (2008)	467
IEEE Xplore database	(Document Title:"Portfolio") OR Document Title:"Multi project") OR Document title:"Multiproject") OR Document title:" Project selection")	–2019	(8) Baker and Pound (1964) (9) Schmidt and Freeland (1992)	209
Project Management Journal® database	(Document title: "portfolio")	–2019	(10) Müller et al. (2008)	147
Back-tracking and Google Scholar	Backtracking of references in Baker and Pound (1964)	1950–1964	(1) Markowitz (1952)	38
Total			The search results include all 10 articles from our test sample	1,970

the abstracts of the 669 articles and disassembling the literature in (1) its basic concepts and (2) time periods. First, we identified and organized concepts into categories and subcategories. In order to do so, we utilized a concept map to display and refine the results of our review (Daley, 2004). As our reading of articles progressed, the concept map's concepts and categories emerged, evolved, and merged. Figure 1, in the next section, presents the final concept map with seven overall categories. Second, to disassemble the literature in a time dimension (i.e., to identify and display when concepts and categories emerge; see Figure 2), each article was coded with the publication year and with the categories that we determined to be the most predominant in the particular article. As some articles touched on more than one category, we found it most accurate to relate each contribution to one to three categories.

The *reconstructing* was performed by putting the disassembled literature together by connecting content and time by starting writing pieces of text. Gradually, a historical narrative emerged, and its grounding in data—the disassembled literature—should ensure its trustworthiness (Gill et al., 2018). Finally, the results of our narrative review were displayed by a visual representation—a model—making it easier for academics and practitioners to engage with our findings (Van de Ven, 2007).

Deconstructing the Literature

As illustrated in Figure 1, our concept map identifies seven overarching categories in the PPM literature.

Starting from the left and moving toward the right side of the figure, we briefly introduce the seven categories. The first category *(1) Portfolio selection and optimization*, in its simplest version, focuses on PPM decisions. It aims to find the optimal mix of projects that maximizes the use of resources and reduces the project portfolio risk. *(2) Management of project ideas* deals with problems and challenges in getting ideas to market and how to secure control and efficiency. We see this as a category of concepts that enable and support the selection process, and thus distinct. *(3) From the fuzzy portfolio front end to decisions and its ongoing implementation* highlights ways to understand and manage PPM as a sequence of decisions from the fuzzy portfolio front end to the project completion. *(4) Governance, maturity, and conceptual models* understands PPM as a system where best practices from maturity and concept models or more abstract governance and governmentality principles may integrate the PPM as a coherent arrangement. *(5) Strategy-oriented approaches* refers to the content of the strategic dimension, and value serves as an integrating term,

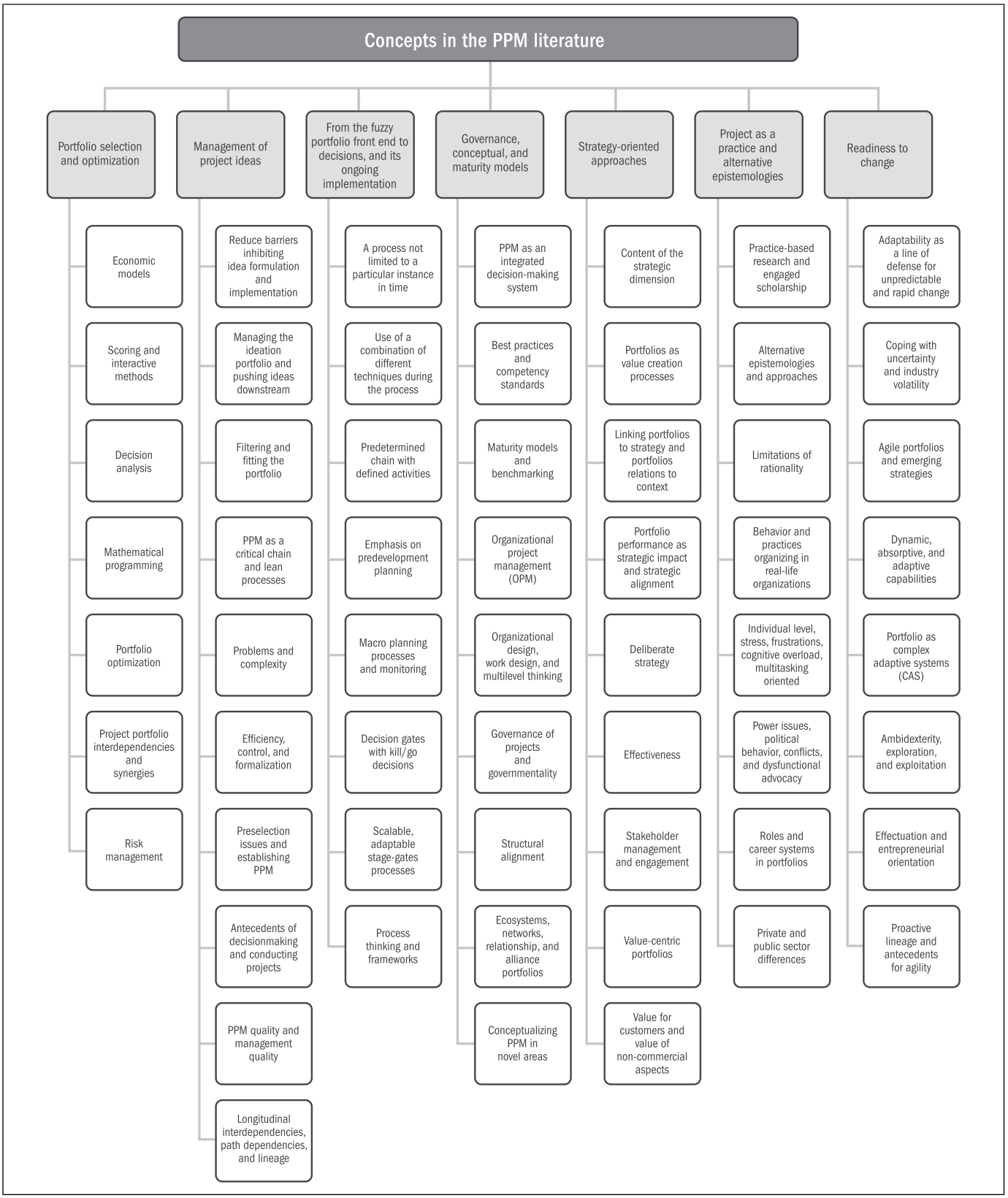

Figure 1. Concept map of the PPM literature.

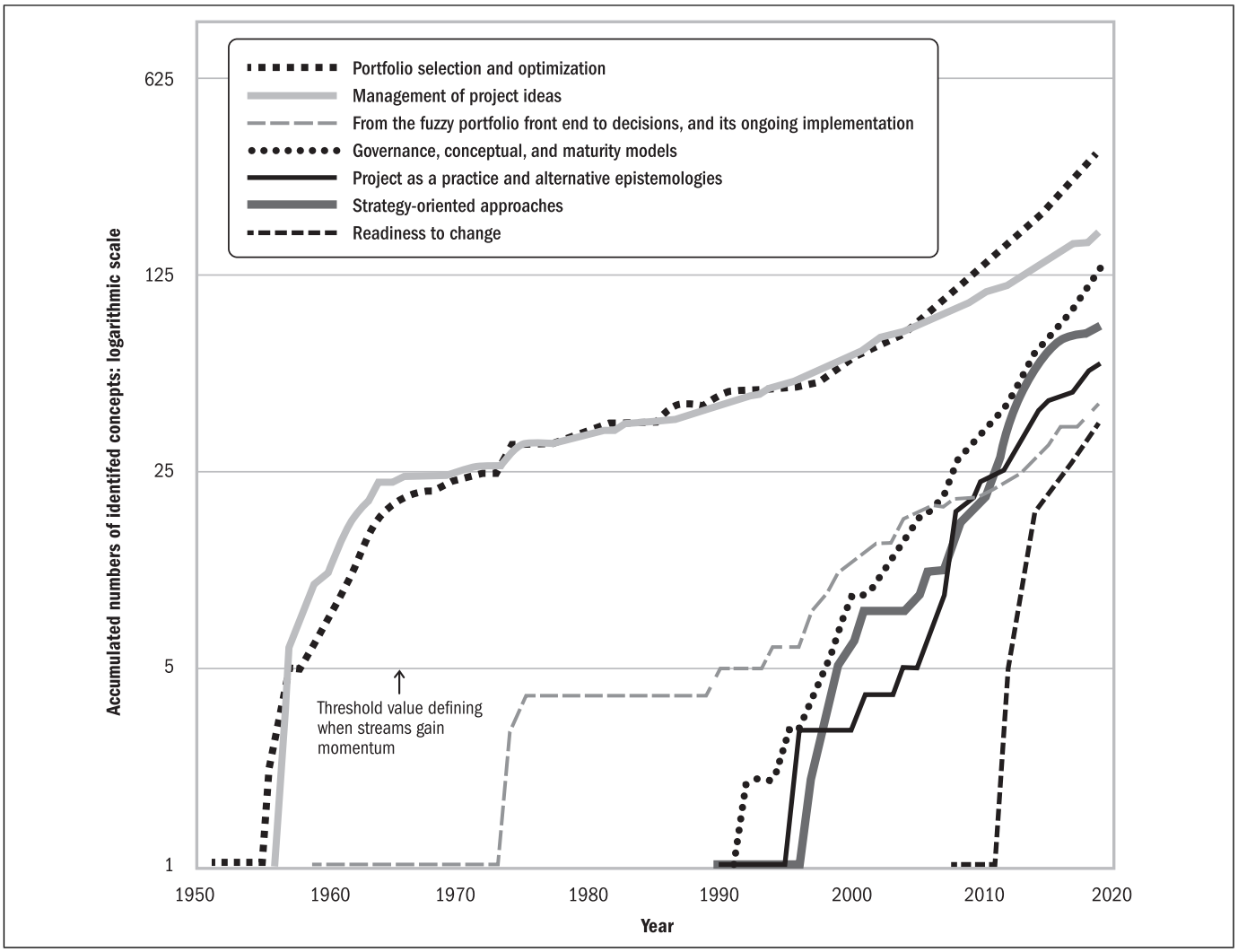

Figure 2. Seven decades of PPM research: How steams of literature emerge and evolve.

for example, by understanding PPM as a value creation process. *(6) Projects as a practice and alternative epistemologies* expands the existing research, namely by addressing the limitations of rationally based models. It turns the emphasis to real-life practices and draws on alternative ontologies and epistemologies. *(7) Readiness to change* refers to the processes of strategy development and implementation and the capabilities required in unpredictable, rapidly changing environments.

Streams of Literature

Concepts in the literature are not static; they evolve in interaction with their historical context (Cummings & Bridgman, 2011). Over time, concepts and categories emerge, gain momentum, reach a point of saturation, revitalize (Sibbet, 1997), or take new directions (Nobelius, 2004). To emphasize our dynamic understanding of the concepts in the seven categories, we hereafter denote these as streams of literature or just streams. Figure 2 presents the accumulation of observed concepts affiliated with each stream across the seven decades of interest in our stock of literature. We define the point in time where the streams have five observations of associated concepts as the point in time where they gain *momentum* in the discipline. See the horizontal line in Figure 2 marking five observations on the logarithmic and accumulated scale on the vertical axis.

Reconstructing the Literature

To establish a chronological structure for our historical narrative, we use the points in time, where each of the seven streams gained *momentum* as a starting point to discuss their origin, content, development, and particular historical context.

Portfolio Selection and Optimization

Heavily inspired by the breakthroughs in portfolio theory in the post–World War II period (Markowitz, 1952; Roy, 1952;

Table 2. Seven Portfolio Selection and Optimizing Techniques

Portfolio Selection Techniques	Examples
Interactive methods	Delphi-aided programming, (Khorramshahgol & Gousty, 1986), Q-sort to achieve group consensus (Souder, 1975), group problem-solving approach (Davies & Pearson, 1980)
Scoring	Scoring model design (Moore & Baker, 1969), scoring and profitability models for evaluating and selecting engineering projects (Dean & Nishry, 1965)
Portfolio optimization	Multi-objective model (Ringuest & Graves, 1989), strategic orientation in decision criteria (Jiang & Klein, 1999), optimal risk sharing (Cadenillas et al., 2007)
Decision analysis	Analytic hierarchy process (AHP) (Meade & Presley, 2002), decision trees in portfolio selection (Tucker & Kim, 2009), multiple criteria decision analysis (Angelou & Economides, 2008)
Mathematical programming	Linear programming (Ringuest & Graves, 1990), nonlinear programming (Schmidt, 1993), dynamic programming (Hess, 1962)
Economic models	Real options (Bardhan et al., 2004), net present value in portfolio selection (Rzasa et al., 1990), portfolio risk diversification (Paquin et al., 2015)
Decision support systems (DSS)/artificial intelligence (AI)	Mini computer to facilitate portfolio decisions (Mandakovic & Souder, 1990), decision support systems (Ghasemzadeh & Archer, 2000), artificial neural network model (Costantino et al., 2015)

Tobin, 1958), the stream of portfolio selection and optimization gained its momentum in the 1950s. Rosen (1956) asked how one picks the right projects, and this question has taken the focus, not only for this stream of literature, but for the entire PPM discipline, for decades—at least when it comes to the number of published articles on the topic. Drawing on Henriksen and Traynor (1999), these contributions on portfolio selection and optimizing can be associated with one of seven overall techniques: interactive methods, scoring, portfolio optimization, decision analysis, mathematical programming, economic models, and decision support systems (DSS)/artificial intelligence (AI). Table 2 presents examples from selected articles.

As early as in the 1960s, scholars began to discuss the limitations of formalized and mathematical based portfolio selection models. While their sophisticated arguments and calculations are elegant, their usefulness is often questioned (Gear, 1974) and their complexity often necessitates the assistance of an expert to be usable by real-world managers (Henriksen & Traynor, 1999). It has been an ongoing challenge for this stream to include all contingencies and to lower the uncertainty of estimates for input (Meifort, 2016). Thus, scholars increasingly aim to include more explanatory variables and are starting to theorize about how different types of selection models fit different contexts (Chaparro et al., 2019). Also, as the knowledge base expands, a broader understanding of selections and optimizations is applied. In that regard, the stream gradually has included more management-oriented concepts supporting the selection processes. For instance, we find a vast amount of articles on risk management (Teller & Kock, 2013; Teller et al., 2014), multiproject interdependencies (Killen, 2017; Killen & Kjaer, 2012), and resource allocation (Engwall & Jerbrant, 2003; Hollister & Watkins, 2018).

Management of Project Ideas

In parallel with the abovementioned stream, a stream of literature defined as *management of project ideas* gained momentum in the mid-1950s. This stream should be seen in light of the high optimism after World War II; most products were sold, and new industries emerged and evolved rapidly. Thus, there was a need for R&D laboratories to come up with new ideas (Rubenstein, 1957b) and to push the ideas and technology downstream toward the marketplace (Nobelius, 2004; Rubenstein, 1957a). One contemporary scholar writes: "There is no standing still in this respect — to stand still is to move backward, with an attendant decline in sales" (J. Adams et al., 1957, p. 40). As this stream of literature develops, it increasingly becomes more theoretical, partially inspired by operations research (Gould, 1960; Johnson & Turner, 1956). Keywords, such as fitting (Norden, 1958), feasibility (Schweyer, 1961), monitoring (Hardingham, 1970), critical chains (Ordoñez et al., 2019), and later the concept of efficiency provided a more overarching term (Martinsuo & Lehtonen, 2007). This establishes the intellectual legacy for some of the most popular PPM topics of discipline, with the common interest of making PPM more efficient and successful. These contributions deal with issues such as solving PPM problems (Elonen & Artto, 2003; Hansen & Kræmmergaard, 2013), improving control (Canonico & Söderlund, 2010; Müller et al., 2008), management quality (Jonas et al., 2013), and formalization (Teller et al., 2012). In recent years, we see that this stream has expanded by moving further upstream as it puts more emphasis on the preconditions for PPM success (Heising, 2012). Such preconditions comprise concepts as antecedents (Kester et al., 2014), ex-ante coordination (Maniak & Midler, 2014), and longitudinal interdependencies (Kock & Gemünden, 2019).

From the Fuzzy Portfolio Front End to Decisions and its Ongoing Implementation

The intellectual legacy of this stream can be seen in the context of the economic crisis in the mid-1970s and the 1980s with shrinking market demands. According to Nobelius (2004, p. 371), this increased the interest for ways to control and reduce the use of resources. Thus, studies searched for new

ways to eliminate wasteful efforts by reviewing and improving how organizations select and monitor projects (Cooper, 1981). Scholars began to consider PPM, not only as a particular occurrence in time, but as a set of ongoing sequential processes where a broad combination of techniques should be applied (Pearson, 1974). The stage-gate model emerged from this thinking, mainly developed and made commercially available by Robert Cooper and his colleagues. Stage-gate models suggest a series of stages, where the project teams undertaking the work provide the required information, followed by decision gates. At each gate, the organization evaluates and decides if the project should continue—a go/kill decision (Cooper, 2008). The principles from stage-gate models have become the default in many companies, and the popularity of this stream of literature peaked in the 1990s and early 2000s, as this period produced some of the most cited contributions in the PPM literature (Cooper, 1990). After 2000, this stream of literature became saturated, and scholars began to question what would be next (Cooper, 2008). In recent years, the basic ideas in stage-gate models have remained unchanged, but there is a focus on making these models more scalable, adaptable (Cooper, 2014), and supportive of agile practices (Cooper, 2016).

Governance, Conceptual, and Maturity Models

In the 1990s, scholars began to advocate for broadening the view of PPM from merely being a sequence of decision events to understanding PPM as an integrated decision system (Schmidt & Freeland, 1992). Scholars developed frameworks linking the entire cycle of PPM as a cohesive system of strategy, stakeholder decisions, methodology, projects, and data (Archer & Ghasemzadeh, 1999). A bit later, in the period after 2000, frameworks of so-called best practices started to gain a foothold (De Reyck et al., 2005; Jeffery & Leliveld, 2004). These best practices allowed individuals and organizations to evaluate the maturity of various aspects against defined standards (Vukomanović et al., 2016; Young & Conboy, 2013). Research has provided some empirical support for a connection between adopting best practices and positive organizational effects (De Reyck et al., 2005; Jeffery & Leliveld, 2004). However, other researchers are more skeptical of using best practices; one substantial criticism is the implicit and dubious notion that one universal and static maturity model fits all organizational settings across time and space (Hansen & Kræmmergaard, 2014; P. Morris, 2013). Such criticism creates research opportunities for more empirically oriented directions, such as looking at PPM as a system connected by organizational governance (Müller et al., 2014; Müller et al., 2017), structural alignment (Kaiser et al., 2015), or work design (Hansen et al., 2017; Petro & Gardiner, 2015). Most recently, scholars have begun to consider PPM as embedded in a model or system consisting of organizational project management (OPM), which complements projects, programs, portfolios, governance, the project management office (PMO), and organizational design (Müller et al., 2019).

Strategy-Oriented Approaches

Around 2000, a stream of strategy-oriented literature gained momentum and scholars began to view PPM performance in light of its strategic impact (Meskendahl, 2010). Earlier research had focused on the *efficiency* of PPM as the principal goal, for example, by reducing project-related problems and improving project performance (De Reyck et al., 2005; Martinsuo & Lehtonen, 2007). Now, scholars address the need to expand this understanding (Müller et al., 2008) by bringing forward the term *effectiveness* (Patanakul & Milosevic, 2009). The term contains the classical operational attribute of PPM (i.e., *efficiency*) but also a broader ability of PPM to align with and obtain strategic goals and obtain the expected value (Patanakul, 2015). Martinsuo and Killen (2014) suggest the concept of strategic value, which encompasses ecological, social, health and safety, societal influence, learning and knowledge development, longer-term business value, and real options (Brook & Pagnanelli, 2014; Maniak et al., 2014). Managing value creation in portfolios appears to be a fruitful path for PPM research (Laursen & Svejvig, 2016), as the notion of PPM value and success has gradually changed focus from being project- and product-centric toward being value-centric (Ang & Biesenthal, 2017). As we expand our understanding of value and success (Laursen & Killen, 2019), we become better at assisting organizations in establishing a more strategic orientation to their external and internal organizational environments (Martinsuo & Geraldi, 2020).

Projects as a Practice and Alternative Epistemologies

In the 2000s, a stream of contributions advocating for more practical-oriented PPM research and the use of alternative epistemologies gained momentum. By applying alternative epistemologies (e.g., interpretative approaches), scholars demonstrate how PPM decisions and processes are driven by more than rational motives (Blichfeldt & Eskerod, 2008; Christiansen & Varnes, 2008; Jerbrant & Karrbom Gustavsson, 2013). Furthermore, by applying empirically driven research approaches, scholars shed light on the roles, practices, and ways to organize PPM in real-life organizations, such as the roles and practices of the middle managers (Blomquist & Müller, 2006), the internal stakeholders (Beringer et al., 2013), the steering committee (Mosavi, 2014), experienced project managers (Gemünden et al., 2018), and the role of the PMO (Bredillet et al., 2018a, 2018b; Unger et al., 2012). Notwithstanding the valuable work that has been done in this stream, we observe calls for more practice-based research (Clegg et al., 2018), human-centricity (Gemünden et al., 2018), and calls for applying less used epistemologies in PPM-related research (Geraldi & Söderlund, 2016, 2018). In particular, we find a surprisingly low number of contributions using the concept of power issues (Nielsen & Pedersen, 2014; Platje & Seidel, 1993), despite the fact that portfolios are arenas with potential conflict between the permanent and the temporary organization (Jonas, 2010). Furthermore, studies show that political issues, for instance, stakeholder influence and competing

behavior, can negatively impact PPM performance (Beringer et al., 2012; Beringer et al., 2013).

Readiness to Change

In the 2010s, scholars began to emphasize that uncertainty in the organizational environment was increasing (Korhonen et al., 2014; Martinsuo, 2013; Martinsuo et al., 2014; Petit, 2012) as more organizations entered the global innovation economy (Feiler & Teece, 2014). This trend motivated PPM researchers to pay more attention to organizational adaptability as a way to cope with unpredictable and rapid change (Kopmann et al., 2017). Thus, researchers started to ponder how to develop adaptability (Biedenbach & Müller, 2012), preparedness for the future (Kock & Gemünden, 2016; Rank et al., 2015), and agility outside the project domain (Stettina & Hörz, 2015). Different theories have been suggested, for example, theory on dynamic capabilities (Daniel et al., 2014; Killen & Hunt, 2010; Killen et al., 2012) and complex adaptive systems (CAS) (Sweetman & Conboy, 2018). Furthermore, practices and theories on agility from research on software development have been diffused to related disciplines (Conforto et al., 2014; Niederman et al., 2018; Petit, 2012) and are likely to be the next frontier in PPM-related research (Dingsøyr et al., 2019; Dingsøyr et al., 2018). As theories on how to improve adaptability and agility begin to be established (Niederman et al., 2018), scholars theorize on how to integrate and balance this adaptability according to existing organizational structures and demands. Thus far, only a few contributions have theorized on this issue, such as the theory on ambidexterity (Eriksson, 2013; Midler et al., 2019; Petro, 2019). Hoffmann et al. (2020) suggest using activity theory to cope with the tension between the demand for being adaptable yet efficient and aligning with strategy. They call for more knowledge on how to identify and deal with the strains mentioned above, as the transition to agile portfolios is assumed to provide challenges to firms due to changes in the existing work structures and mindsets of employees and management.

Discussion

According to Peter F. Drucker and his colleagues, it is futile to predict the future in human affairs. Instead, it is possible to prepare for the future by evaluating what has already happened, as the challenges for the next decades have already taken shape (Drucker et al., 1997). In that regard, a model is a helpful way to visualize patterns in complex data (Van de Ven, 2007) and to integrate the results of a narrative review (Klein & Müller, 2020). Following the argument from Svejvig and Andersen (2015), we assume that the knowledge of the discipline accumulates over time. Broadly inspired by Müller et al. (2019), we propose an onion model that displays the discipline's evolution, and the emergence of the identified literature streams adds

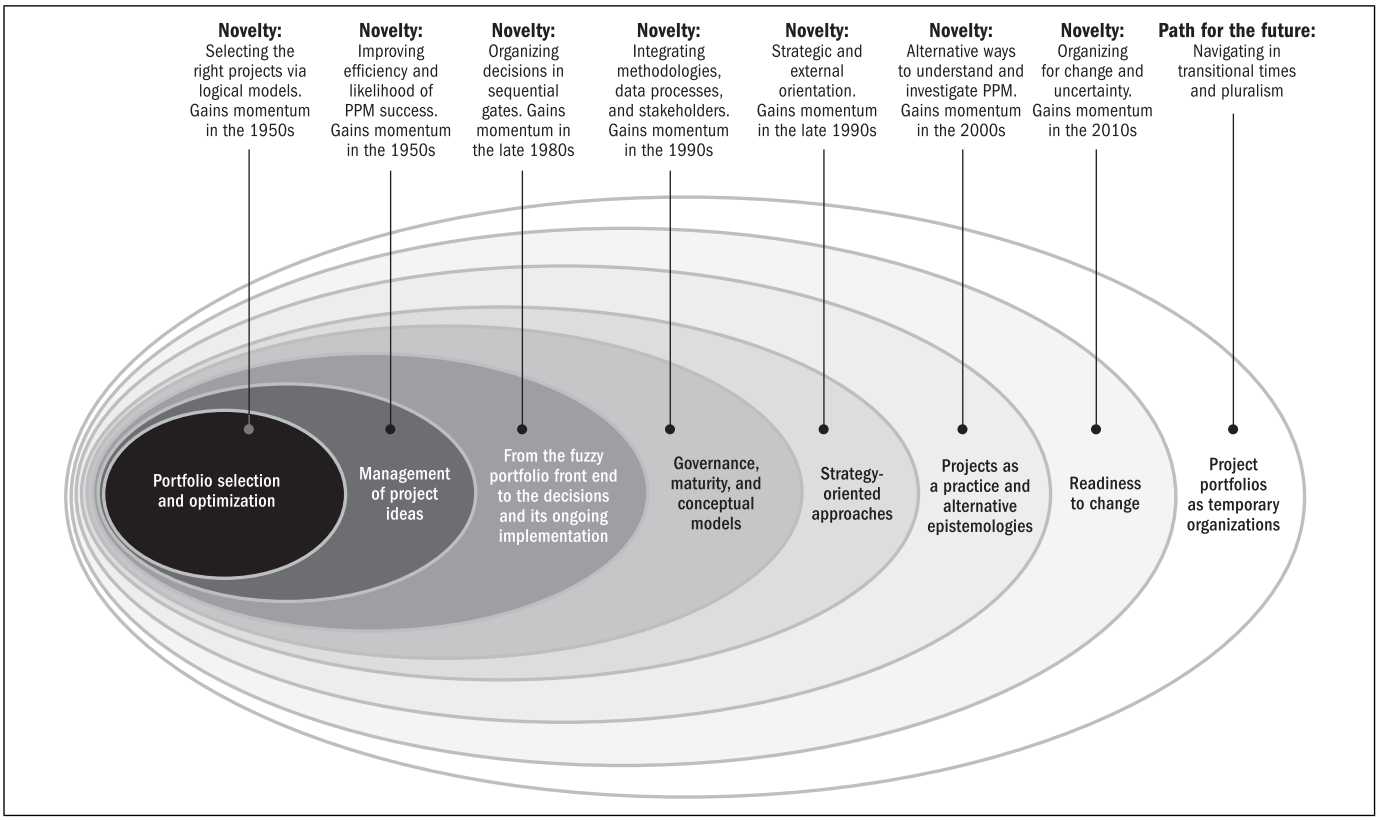

Figure 3. The onion model of seven decades of PPM knowledge.

Table 3. Changing Aspects of Temporarity in PPM: The Past, the Future, and its Implications

Aspects of Temporarity	In the Past	In the Future	Implications for Future Research
Time	Linear understanding	Cyclical understanding	Focus on agility, adaptability, and emerging strategies
Task	Providing tangible products	Providing services	Focus on relationships embedded in corporate networks or ecosystems
Team	Hierarchical and bureaucratic structures	Autonomous and self-managed teams not restricted to organizational boundaries	Focus downward in organizations to how people translate, improvise, and make sense of PPM in practice
Transition	Permanent structure and fixed goals	Temporary structures reproduced by actions	Focus on how actions produce and reproduce objects, actors, and structures

new layers to the existing knowledge base. An organic onion metaphor also communicates that each layer has continued to grow across the seven decades, as earlier shown in Figure 2. Thus, the findings of our model will most probably shape the future of research for decades to come. Our onion model with its seven established layers—the past—can be seen in Figure 3. Next, we will discuss what could be the onion's next layer, that is, the next agenda of the discipline, by reflecting on the fundamental aspects that have led us to where we are now. Our body of literature puts us in a privileged position to do so (Baumeister & Leary, 1997).

Any analysis of human affairs is seen from a certain point of time and is affected by its historical context thus creating different criteria to appraise what is essential. Following current research, we define our historical point of reference as the area of transitional times, an area characterized by temporarity (Tell et al., 2020). By drawing from Lundin and Soderholm (1995), we discuss how the four aspects of temporarity—*time*, *task*, *team*, and *transition*—have changed in the discipline's past and how they will have decisive implications for future research. This discussion is summarized in Table 3.

First, affected by larger historical proceedings, the discipline's understanding of *time* has increasingly changed from a linear understanding to a more cyclical understanding (Niederman et al., 2018; Sweetman & Conboy, 2018). For instance, the PPM selection models developed in the 1950s are most useful in a static business environment as they extrapolate their predictions based on fixed and defined variables. As we approach a more dynamic, complex, and uncertain environment, a more cyclical understanding of time becomes salient. In particular, we see how the *readiness to change* stream, during the last two decades, has offered concepts to cope with such uncertain environments, for instance, agility, adaptability, dynamic capabilities, and emerging strategies.

Second, the characteristics of the *tasks*—broadly understood as the products performed by organizations—have changed. In economies of developed nations we see a shift from companies as providers of tangible products toward companies as providers of services (Killen et al., 2008). Production of services demands, in general, a more holistic exchange of intangibles, skills and knowledge, and processes across organizational boundaries (Gupta, 2018; Vargo & Lusch, 2004). This change has implications for core elements of the PPM discipline, for example, by making the selection and resource allocation processes more complex and dispersed. Previously, selecting and resource allocation were mostly performed *within* the organizational boundaries (Archer & Ghasemzadeh, 1999). Today, such processes and practices are increasingly externalized and embedded in corporate networks or ecosystems (Hannah & Eisenhardt, 2018; Koch-Ørvad et al., 2019).

Third, the more distributed and dynamic business landscapes with less hierarchical and bureaucratic structures that exist today make self-managed *teams* increasingly important (Khanagha et al., 2021). Thus, research must continuously determine ways to push power, decision-making, and resource allocation lower down in the organization (Ancona et al., 2019). Consequently, this shifts our gaze—from centralized decision-making as proposed by the literature in the *portfolio selection and optimization* stream—downward to how the people involved in PPM translate, improvise, and make sense of PPM in practice (Clegg et al., 2018). As teams and temporary organizing increasingly become central to employees' career paths and working life—this temporary organizing is not necessarily affiliated to one organization's boundaries; thus, new forms of formal and informal contracts (Akkermans et al., 2020; Keegan et al., 2018) and career systems must be designed (Gemünden et al., 2018).

Fourth, *transition* has become vital with the increasing pace of change in business environments, and the last two decades have seen the birth of an unprecedented number of new-to-the-world markets (McDonald & Eisenhardt, 2020). Thus, discovery-driven strategies and portfolios must continuously legitimize themselves by demonstrating value and potential via ongoing experimentation and testing (Thomke & Euchner, 2020). Consequently, organizations more frequently review and adapt their portfolios and structures (Dingsøyr et al., 2019; Sweetman & Conboy, 2018). The PPM discipline has gradually moved away from understanding practices and portfolios as having a static nature, as implicitly suggested by the stage-gate approach (Cooper et al., 1997). Instead, later research suggests understanding PPM as having a more shifting and temporary nature (Martinsuo & Geraldi, 2020; Turner &

Müller, 2003) where *actions* produce, reproduce, and change its objects, actors, and structures (Sergi et al., 2020).

Transition and temporarity also enable us to understand how each layer in our model becomes more or less relevant due to the changing historical context. Our analysis has shown how larger historical proceedings and events have inspired new research streams. For instance, post–World War II optimism and the unsaturated demands of the market in those days were the raison d'être of the *management of project ideas* stream of literature. Thus, current and future historical events and trends, such as the emergence of artificial intelligence (AI), global warming, or pandemics, will motivate researchers to introduce new streams—these streams will probably be shaped by concepts we already know. For example, the COVID-19 pandemic makes what we know about ecosystems and networks critical, as organizations increasingly need to enable employees and collaborators to work from anywhere and anytime (Choudhury et al., 2021). Thus, PPM research has a vital role in bringing forward what we know on attracting employees and managing resources in ecosystems and talent networks (Kuhn et al., 2021).

Our historical study serves as a baseline to evaluate what we already know and reflects on areas for development. Theory on temporarity (i.e., Lundin & Soderholm, 1995) has proven useful to helping us point to the future. In that regard, our results accompany current scholars who suggest defining PPM as a (semi) temporary organization (Martinsuo & Geraldi, 2020). We find this inspiring as it can stimulate research with a novel perspective, but it also reaches back to existing literature streams with untapped potential. For instance, defining PPM as a temporary organization directs our gaze on what real-life organizations do (actions) and thus encourages research in the stream entitled *projects as a practice and alternative epistemologies.* We have found the number of contributions in this stream only to be a fraction of the overall amount of PPM articles (see Figure 2). We argue that this points to a fruitful research opportunity, as PPM literature repeatedly has emphasized the importance of action-oriented, situated, and contextualized knowledge (Clegg et al., 2018; Martinsuo, 2013). In contrast, we find a very high number of contributions in the stream *portfolio selection and optimization.* This stream—the historical core of the PPM discipline—continues, after seven decades, to publish the highest amount of new articles, and it seems likely that it will keep this position in the future (see Figure 2). An ongoing challenge for this stream is to demonstrate the practical applicability of its idealized decision-making theory and models (Blichfeldt & Eskerod, 2008; Clegg et al., 2018). Solving this issue is requisite, because PPM as an applied discipline needs to demonstrate its practical relevance; otherwise, it is in danger of losing legitimacy over time (Van de Ven, 2007).

Overall, our study and model present PPM as a pluralistic discipline with a pragmatic application of its different layers, each having different methodological, theoretical, philosophical assumptions, and managerial implications. We believe that there is a strong relationship between progress in research and such diversity (Söderlund, 2011). Thus, we encourage scholars to continue this expansion of the discipline's knowledge base. However, over time, pluralism can lead to the discipline becoming fragmented. To counter fragmentation, scholars call for strengthening the reflectiveness and awareness of its differences (Martinsuo & Geraldi, 2020; Söderlund, 2004). Our research responds to this call by providing a visual representation of research streams in the discipline and its evolution to assist such scholarly thinking.

Finally, our model may engage practitioners in getting a more accurate picture of a complex reality (Knudsen, 2003). We believe that the combination of our model with multiple layers, its historical awareness, and an informed practice will lead to a more insightful research agenda for the future (Geraldi & Söderlund, 2016; van Marrewijk & Dessing, 2019).

Conclusion

Although PPM has been on the research agenda for at least seven decades, the number of contributions on the topic continues to grow. One major challenge is the complexity of navigating a research discipline that is spread across many fields and a long historical period. Conducting a narrative literature review enables us to integrate and build on what we already know and understand where we came from and where we are going. Thus, our literature review traces the literature back to its origin and builds and synthesizes a historical narrative into a model. By evaluating what has already happened and what will most likely shape the future, we suggest an agenda for the decades to come.

Our study is limited by the applied keywords used to select the 669 articles in our review, as related terms may have provided contributions that could and most likely should have been included. In addition, not including books and so-called gray literature probably excluded relevant contributions. Thus, we encourage scholars to conduct literature reviews that include these contributions, as this may identify significant current and emerging streams of literature not captured by our research. Our suggestions for the future are limited by primarily building on past-future thinking, and thus, it is incremental in its nature. It could be interesting to see more contributions by envisioning the future and, on that basis, suggest what we need to do now. It could also be enlightening to see contributions utilizing progressive theory (e.g., Svejvig & Andersen, 2015) to rethink the discipline's current state.

Despite the aforementioned limitations, we argue that our research provides the following contributions. First, our onion model of the discipline offers a way to understand how the discipline has taken shape and outlines the streams of literature that will most likely shape the future. Second, by visualizing the origin and development of the discipline, we have made it easier to understand the key characteristics of its multiple streams and concepts. Stimulating such reflectiveness helps the discipline from becoming fragmented and informs

discussions on how we can exploit existing knowledge as well as cultivate new streams of research. Third, the results of our historical analysis sharpen our vision of the present and create a baseline to evaluate how new trends differ from what we already know. Fourth, we believe that our study supports a collective identity of the discipline and supports more insightful, creative practice and research. Finally, we contribute to the ongoing strengthening of the discipline by suggesting a research agenda that opens a new avenue by building on what we already know.

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Appendix A

Field of Research	Literature Review or Conceptual Paper
Research and development (R&D)	Baker and Pound (1964)
	Baker (1974)
	Schmidt and Freeland (1992)
New product development (NPD)	Brown and Eisenhardt (1995)
	Nobelius (2004)
	Killen et al. (2007)
Innovation portfolio management (IPM)	Rothwell (1994)
	R. Adams et al. (2006)
	Srivannaboon (2016)
	Meifort (2016)
IT project portfolio management (IT PPM)	Kumar et al. (2008)
	Frey and Buxmann (2012)
	Hansen and Kræmmergaard (2014)
Project portfolio management (PPM)	Payne (1995)
	Eskerod (1996)
	Archer and Ghasemzadeh (1999)
	Killen et al. (2012)
	Martinsuo (2013)
	Clegg et al. (2018)
	Danesh et al. (2018)
	Martinsuo and Geraldi (2020)
	Micán et al. (2020)
Alliance portfolios and business relationship portfolios	Wassmer (2008)
	Tikkanen et al. (2007)

Appendix B

Criteria	Reasoning and Examples for the Inclusion of Publications
Portfolio management of projects of all kinds	This concept is the focal point of our research, as we broadly define PPM as the overall organizational ability to manage project portfolios strategically and holistically to better support the success of the organization (Clegg et al., 2018; Killen & Hunt, 2010). Here we include all types of project portfolios in project organizations with enormous diversity. Some of the less typical examples include PPM for academic libraries (Vinopal, 2012) or PPM selection in the sustainable mining industry (Pimentel et al., 2016).
Model and technique applied on project portfolios	Most of the contributions conforming to this criterion are designed to support portfolio selection, and some of the present examples were included in the review. For example, the publication by Montajabiha et al. (2017) proposes an algorithm for project portfolio selection using real options valuation, and the publication by Mohagheghi et al. (2017) suggests a mathematical modeling approach for high and new technology project portfolio selection in uncertain environments.
Project selection processes	This concept includes the entire selection process. Publications utilizing this concept typically investigate how the selection of projects connects to other organizational areas and have mostly been published some decades ago (Pearson, 1974). An example of an often cited publication is Archer and Ghasemzadeh (1999).
Project studies	The project studies approach applies overarching definitions of various levels of analysis. In relation to our selection of relevant publications, the focus of this concept varies, as do its implications for PPM. For example, we chose to include the publication by Geraldi and Söderlund (2018) as it provides explicit implications for PPM research by showing how project studies may include the portfolio level, coined in the publication as "macro project studies."
Governance of projects	The concept *governance of projects* is sometimes just defined as governance. Müller et al. (2014) describe governance of projects as the collective project governance, thus referring to program and portfolio level issues. Applying this definition, we include the paper by Sirisomboonsuk et al. (2018) as it provides insights into our subject matter. This concept easily gets confused with *project governance*, which deals with project-level issues (Müller et al., 2014), and such papers have been excluded from the review.
PMO (project management office)	The term project management office (PMO) can be understood as the support office, often for a large project or the unit fulfilling a controlling, supporting, or coordinating role for one or more portfolios (Unger et al., 2012). As we adopt the latter definition, we find the concept of PMO to be central to PPM research and include publications such as Bredillet et al. (2018a) and Bredillet et al. (2018b).
Papers with explicit implications for PPM	Publications with explicit implications for the PPM level were included, for example, the literature reviews by Kwak and Anbari (2009) and Laursen and Svejvig (2016). Nevertheless, our pool of publications also includes contributions that do not mention the word "portfolio," but still have implications for PPM.
Papers about PPM, but using a different wording	This criterion is designed to capture the older publications before the discipline started to use the word portfolio, but clearly dealing with PPM-related matters. For example, papers using wording as "selection of projects" (Mottley & Newton, 1959), "how to pick the best projects" (Rosen, 1956), and "selection and evaluation of R&D projects" (Hitchcock, 1963).